# N83 one-target IC vs strong rho: Tier 1 protocol

2026-10-14

## N83 one-target IC against strong rho: Tier 1 protocol

**Status (2026-10-14): protocol frozen and validated at reduced scale. The Tier 1 run has not happened, so this directory contains no timing result.** The protocol re-measures the n=83 a=1 one-target comparison on a fresh GitHub Actions `ubuntu-24.04` instance. Today that comparison rests on three unisolated packed-rho pairs with a median rho/IC wall ratio of 8.2× (crypto experiment). This protocol is the n=83 step that the crypto `vs_rho` ledger row names as next: "extend the isolated-host protocol to n=71 and n=83 with the same five-phase interval and taskset receipt". It uses the same producers, claim builder and claim checker as the n=61 isolated rung. Each claim is reported with its cold ratio and the Bernstein–Lange model column, as ledger §23 requires.

| Item | Value |
|---|---|
| Curve | `EC1N83Ce1h35c7ced6b497` (`icv1-f2m83-t6151469093347-9eba6cd5`): $y^2 + xy = x^3 + x^2 + 1$ over GF(2)[z]/($z^{83}$+$z^7$+$z^4$+$z^2$+1) |
| Subgroup | r = 8569786107849059 (2^52.93), cofactor 1128547018, signed-Frobenius action of 166 elements |
| Factor base | retained K600: 600 orbit columns, 99,600 points, base hash `d4a4da26...a9f7` |
| Frozen point Q0 | (265589834553716618053296 0, 3881526307991673265292615); its validation scalar is opened only by the verifier |
| Crypto source | `770f6a6925a0d29b768bd79074f54a173e967fe7` (on crypto `origin/main`), restored path-by-path with `git archive` and hash-checked |
| Harness | crypto beat `koblitz.compact_orbit.n83_single_target`; `single_target_panel.py` claim builder; `boundary_autolab.validate_claim(stage="vs_rho")` |

### What changed from the 2026-10-06 runs, and why

1. **The rho reference.** The 8.2× runs used `koblitz_rho_fixture ... packed`. This build refuses that backend as a `vs_rho` reference: it stores every step and canonicalizes with an O(n) scan. The crypto evaluation of 2026-10-12 (`research/n83_evaluation_20261012/`) measured the same Q0 against default strong rho. Its paired IC/strong online geometric ratio was 1.38 [1.04, 1.88], meaning strong rho was faster online. The primary arm here is strong rho rung 3 (`koblitz_rho_batch_ks_strong_online`: 32 lanes, 4 distinguished-point bits). This is the n=61 rung's rho, and its wide path reaches n=83. Panel C re-runs the legacy packed arm under isolation only to test whether the historical 8.2× survives.
2. **Where the setup boundary falls** (the gap the README names). The rho interval that was kept included setup whose target dependence was never established. The pinned sources settle it:
   - Strong rung 3 draws its 32 jumps as multiples of G alone and starts every walk at [c]G + Q. Before `after_target_built`, the only work on Q is parsing it. Curve construction, the normal basis and the jump table are therefore target-independent. They are logged per process as `online_start_ns` and excluded from online time.

- The packed backend builds its jumps as aG + bQ, so its 7.5 ms jump setup depends on Q and correctly sits inside its `total_ms`. Panel C reports it separately as `setup_ms`. The packed run's curve construction was untimed and outside the interval.

3. **The IC precompute.** Every arm on the runner is pinned to one CPU. Running the rank stage inside each IC arm would cost about 15–25 minutes per process, which does not fit eight pairs in 6 hours. `ic_log_cache.patch` adds two environment variables to the wide path of the online producer. `KIC_LOG_TABLE_OUT` writes the 600 factor logs after an in-process rank stage. `KIC_LOG_TABLE_IN` skips the rank stage and loads the table. It checks n, a, r, base hash and width, then verifies every `[L_j]G = R_j` before any target work. The load and verification sit in the relation-LA slot of the reusable setup. The target code and its five phase timers are unchanged. The table is reproducible and hash-bound because `L_j = log_G(R_j)` depends on the base alone; the rank seed only changes how it is found. The rank stage runs once per job on the same pinned CPU. It is timed, logged, and excluded from online time. It is charged in the cold ratio and in the BL budget. `precompute.py verify` re-checks all 600 logs in the harness's pure-Python `oracle.py`.

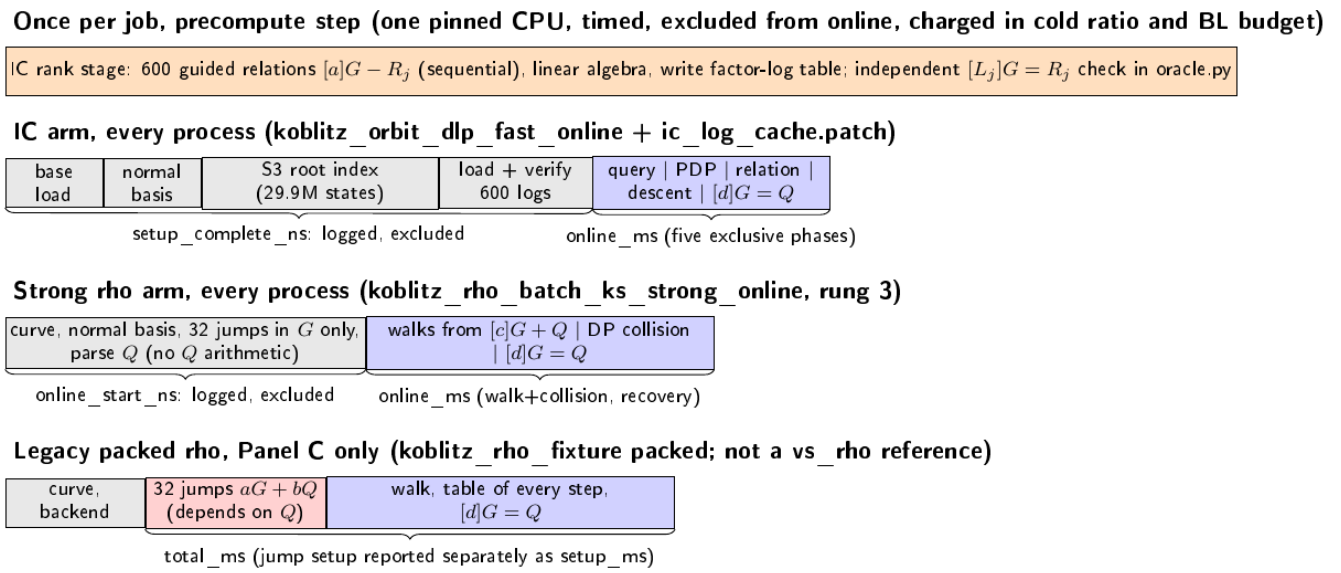



Figure 1: Clock boundaries

*Figure 1. Clock boundaries (source, vector). Grey: target-independent setup measured in every process, logged and excluded. Orange: the once-per-job IC rank stage. Blue: online intervals. Red: the packed backend's Q-dependent jump setup, inside its interval.*

## Frozen design (`tier1_protocol.json`)

| Panel | Role | Targets | Pairs | Rho arm and seed | Statistic |
|---|---|---|---|---|---|
| A | primary | frozen Q0 | 8, alternating RC/CR | strong rung 3, seed 831000 (not run before freezing) | median of within-pair rho/IC ratios; isolated_bench `tier1_bootstrap95_by_pai` (10,000 resamples, seed 20261011); paired geometric mean as secondary |

| Panel | Role | Targets | Pairs | Rho arm and seed | Statistic |
|---|---|---|---|---|---|
| B | §23 over targets | the beat's first 16 public hash-to-curve points (`kic-n83-single-target-public-point-v1`, role `eval`) | 16, one per point, order alternating | strong rung 3, seed 830000 + w | §23: mean rho online / mean IC online, bootstrap over targets (10,000, seed 2026101400) |
| C | legacy | frozen Q0 | 8, alternating | packed, seed 202610061 (historical R1, deterministic 6,608,900 steps) | as A; not a `vs_rho` reference |

- **Envelope:** `scripts/isolated_bench.py run` launches every arm in its own process, pinned to one execution CPU with `RAYON_NUM_THREADS=1`. The isolated_bench Tier 1 gates apply: preflight, no change in `nr_throttled`, `oom` or `oom_kill`, an unchanged binary, exit 0, `verified=1`, and stable artifacts. Pair fields (`panel`, `target_sha256`, `recovered_scalar`, `curve_id`, `arm_verified`) must match within each pair, so both arms must recover the same scalar on the same point.
- **Claims:** for every Panel A and B pair, `verify_tier1_receipt.py` builds the n=61 harness claim (IC1 candidate, workload and run IDs, five IC phases, rho walk/collision/recovery) and replays both certificates in `oracle.py`. It then runs `validate_claim(stage="vs_rho")`.
- **Reported beside every online figure:** S = native ops / $\sqrt{r}$, with IC root-index probes and rho walk steps kept as separate units, since no common unit is claimed. Also reported: rho steps against the floor $\sqrt{(\pi r/4n)}$; IC online priced in measured rho steps; the cold ratio (per-process setup + rank stage + online, IC over rho); the break-even target count; and the BL model column, IC online over $1.93{\cdot}1.21{\cdot}r{\cdot}c^2/(2n{\cdot}P)$. In that formula c is the measured rho ms per step and P is the IC setup.
- **Decision rule:** a panel is eligible only if every pair is valid, every claim check passes, both certificates replay, both arms agree, and Q0's scalar equals the sidecar. On an eligible panel, an interval lower bound above 1 reads "IC online faster", an upper bound below 1 reads "rho online faster", and an interval containing 1 reads "not separated". Any ineligible panel is "no result". Panel A's reading is the instance-scoped Tier 1 result for Q0. It is always quoted with Panel B, the cold ratio and the BL column, and never as a speed of the method. No K or seed is re-chosen after seeing a result.

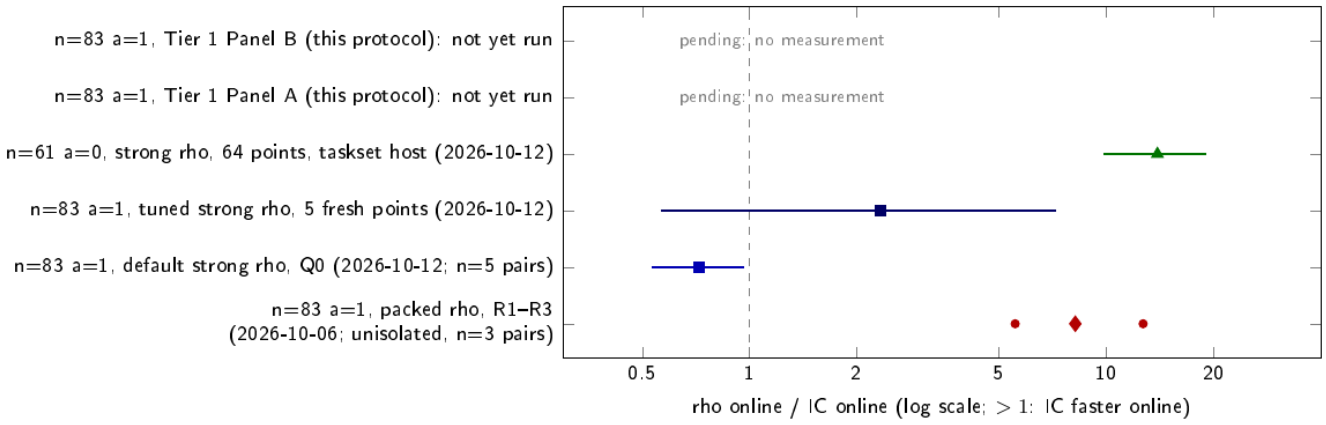



Figure 2: Prior ratios

*Figure 2. Prior rho/IC online ratios that this protocol re-measures (source, vector). Values are as recorded in the crypto `vs_rho` row, the 2026-10-06 runs and `research/n83_evaluation_20261012/`; intervals are 95%. The tuned-rho row used one shared IC setup for five targets. The two Tier 1 rows are empty because the run has not happened.*

## Resources and expected runtime

| Stage | Estimate | Basis |
|---|---|---|
| Build + example tests | 5–15 min | 5 m 30 s with `-j 2` here; 19 example tests in 29 s |
| Rank stage, one thread | 15–35 min | 1.37–1.49e9 probes; 1.63e6 probes/s single-thread here (n=65 K200 rank); 1,048 s for all single-worker IC preparation on the 2026-10-12 Linux VM |
| Panel A, 8 pairs | 13–27 min | per IC process: index 65–100 s + online 6–13 s; strong rho 3–6 s |
| Panel B, 16 pairs | 27–53 min | same per pair |
| Panel C, 8 pairs | 18–40 min | packed rho 37–71 s per process |
| Claims, replay, audit | 2–10 min | `oracle.py` curve set-up 4 s; 48 certificates |
| **Total** | **about 1.3–3.0 h** | job timeout 350 min, inside the 6 h limit |

Memory: the IC root index peaked at 4.4 GB in the 2026-10-06 target receipts and at 5.75–5.81 GB including the rank stage. Packed rho peaked at 1.1 GB and strong rho at about 120 MB. Every arm stays inside 16 GB, and the panels run one process at a time.

The frozen `koblitz_orbit_dlp_fast` precompute took 4.8 min with 12 threads on Apple arm64, 3,660 thread-seconds if it scaled linearly. With 4 threads on the runner's 2 SMT cores, that parallel path would take about 5–15 min. It is not used here: the online producer has only the sequential rank, and one pinned CPU keeps the rank stage in the same unit as the arms. On peak memory, 4 threads should stay at or below the 12-thread 5.75 GB, since the root index dominates.

**Committing the log table.** No committed table exists yet: `committed_log_table_sha256` is `null`. Computing the n=83 table in this container would have meant a 5–6 GB, 15+ minute job on a shared container with 1.5 GB of free disk, so the smoke test ran at reduced scale instead. After the first Tier 1 run, the archived table (about 25 KB) should be committed under `inputs/` and its SHA-256 bound in the protocol in a separate change. Later runs then check that they reproduce it rather than trusting it.

## Validation done here (reduced scale; not timing evidence)

The container is shared and contended, and the clocks quoted below are diagnostics only. Receipts are in `smoke/`.

| Check | Result |
|---|---|
| `restore_crypto_source.py` | 18 file digests, the patch digest, the Rust tree digest and the snapshot tree digest match the protocol |
| `cargo build --release --locked` | pass; `cargo test --release --locked --bins`: 3 + 8 + 8 = 19 passed |
| Strong rho, n=83, Q0, seed 202610141 | recovered 3141592653589793, the validation sidecar; 4,282,875 steps; verified |
| Packed rho, n=83, Q0, seed 202610061 | recovered the same scalar; 6,608,900 steps, identical to archived R1 |
| IC wide path, n=65 a=0, K200 (26,000 points), rank in process | 200/200 relations, 0 failures; table written and `[L_j]G=R_j` asserted |
| IC with `KIC_LOG_TABLE_IN`, same base, public point | recovered 123456789012 with the identical relation (point indices, 3,621,628 probes); rank skipped |
| Strong rho, n=65, same point | same scalar |
| Tampered table (one log + 1) | producer exits 101: "factor log 5 does not map G to its representative"; `oracle.py` check FAIL on column 5 |
| `isolated_bench.py run`, n=65, 2 pairs | 4 valid rows, 2 valid pairs, pair fields matched |
| `isolated_bench.py probe`, Tier 1 manifest | **accepted** (`ok: true`, docker, host exclusivity "unverified"). The Tier 0 probe rejects with 16 problems |

| Check | Result |
|---|---|
| Claim builder + `validate_claim`, n=83 | PASS, all nine replay checks true. Schema check only: it used the archived R1 relation and R1 timers, not a measurement |
| `make_targets.py` | 16 Panel B points derived with the harness law; disjoint from Q0 |
| Workflow | YAML parses; every `run:` block passes `bash -n`; the embedded Python compiles |

## Gaps

- No n=83 IC process ran here: not the online producer, not the patched wide path, not the n=83 rank stage. They run first on the runner. The wide patch was exercised at n=65, and the n=83 IC path was last run in the 2026-10-06/12 evidence.
- The Tier 1 probe accepted this container, contrary to expectation. Tier 1 observes counters and does not certify exclusivity, so its acceptance here shows the gate is weak, not that this container is isolated.
- Panel B's `speedup_claim_eligible` flag from isolated_bench is always false, because its gate needs eight repetitions per case. Panel B's reading uses the protocol's own rule and the §23 statistic instead.
- The precompute and the `KIC_LOG_TABLE_IN` load are not noise-gated; they are instance observations. Cold ratio and BL inherit that.
- Deviations from the crypto beat: K is fixed at 600 with the retained base (no {200, 600} tune), the factor-log table is cached, and Panel A is added. Results are instance-scoped. Independent-host replay and ledger promotion stay outstanding.
- Canonical graphs: none changed, because there is no new measured fact. Checked: `experiments/volcano-ic/figures/` (census, ecdlp and tau figures; no n=83 vs_rho data) and crypto `docs/index-calculus-scoreboard.html`, whose `n83-retained-gates-20261010` entry concerns the a=0 81-bit model and stage-only gates. The crypto ledger row, scoreboard and `benchmarks/` entry belong to the PR that lands the Tier 1 result.

## Files

`tier1_protocol.json` (frozen design and digests), `restore_crypto_source.py`, `ic_log_cache.patch`, `Cargo.toml`/`Cargo.lock`, `make_targets.py` and `inputs/targets.json`, `precompute.py`, `arm.py` (the per-arm wrapper isolated_bench launches), `make_tier1_manifest.py`, `verify_tier1_receipt.py`, `figures/`, `smoke/`, the workflow `.github/workflows/n83-one` and the PDF `n83_onetarget_tier1_protocol.pdf`.

## Reproduce

```
git clone https://github.com/aburan28/crypto /path/crypto
python3 restore_crypto_source.py --crypto-repo /path/crypto
python3 make_targets.py --check
cargo build --release --locked
python3 precompute.py run --binary $CARGO_TARGET_DIR/release/ic_online_logcache --cpu 0 --out R/precompute
python3 make_tier1_manifest.py --workdir <repo> --bin-dir $CARGO_TARGET_DIR/release \
  --log-table R/precompute/factor_logs_n83_K600.json --precompute-receipt R/precompute/precompute_receipt.j
  --cpus 0 --execution-cpu 0 --provider P --instance-id I --allocation-receipt R/allocation.json --output-d
python3 ../../scripts/isolated_bench.py --queue-root R/queue run R/manifests/manifest_A.json > R/run-summar
python3 verify_tier1_receipt.py R
```